GraphFuse: Fusing Dense and Graph Retrieval for Multi-Hop QA

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Abstract

Dense and graph-based retrievers fail on different questions. GraphFuse runs a GraphWalker-QA walk and a HopRank chain search in parallel and merges their candidate chains with a learned fusion scorer. GraphFuse reaches 57.9 and 46.3 EM on KaveriHop 2-hop and 3-hop, and 41.2 and 22.7 EM on TriHop-Bench 3-hop and 4-hop. Fusing dense and graph retrieval outperforms either approach alone at every hop count.

Keywords: hybrid retrieval, fusion, multi-hop question answering

1 Introduction

Iterative dense retrieval [1] remains the backbone of many systems. HopRank [3] is strongest on short chains, while GraphWalker-QA [2] is strongest on long chains. Their errors overlap on fewer than 40% of questions, which suggests that combining them should help.

2 Method

GraphFuse extends GraphWalker-QA [2] and HopRank [3]. Both components produce up to 50 candidate chains; a fusion scorer combines walk scores, rerank scores and the agreement between the two candidate lists.

3 Experiments

We evaluate on KaveriHop [4] and TriHop-Bench [5]. Baseline numbers are quoted from the original papers and from the TriHop-Bench paper.

Method	Source	KaveriHop 2-hop EM	KaveriHop 3-hop EM	TriHop-Bench 3-hop EM	TriHop-Bench 4-hop EM
GraphWalker-QA	[2]	50.1	39.8	–	–
GraphWalker-QA	[5]	–	–	34.7	17.5
HopRank	[3]	53.4	36.0	–	–
HopRank	[5]	–	–	31.9	11.2
GraphFuse	ours	57.9	46.3	41.2	22.7

Table 1: Exact match on KaveriHop and TriHop-Bench.

4 Conclusion

Fusing dense and graph retrieval outperforms either approach alone at every hop count. Fusion roughly doubles retrieval cost, which we leave to future work.

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